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Development of an Image-based disease detection leveraging AI/ML techniques and Computer Vision



By

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Project Card

Start Date and End Date of the project

Start Date: 01 December 2025
End Date: Ongoing / As per TIH schedule

Total Sanctioned Amount

₹ **2.4**

Lakhs

Total Amount Disbursed from TIH

₹ **96**

Thousand

Awards/Recognition

- Trained model for Soyabean crop
- Tested the Mahavistar App after the deployment
- Trained multiple VLMs

Action Items from the previous review

No previous review taken. Uploaded the PPT as asked. Had review meetings for VLM related discussions.

IP/ Patents details

No IP or patent filings are applicable at this stage.

Commercialization Plan

- Deploment through Mahavistar App for real world agricultural usage.
- Improve scalability and usability for farmers.
- Extend support for additional crops and diseases.
- Support caption-based diagnosis



Background & Problem Statement

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Domain
Agriculture Technology

AI-driven crop health monitoring, precision agriculture, and automated disease diagnosis for improving productivity and reducing losses.

Focus Area
AI/ML + Computer Vision + Vision Language Models

Deep learning, CNN architectures, validation systems, and multimodal models for intelligent plant disease diagnosis.

Core Need

Develop robust systems that provide actionable insights to farmers while ensuring reliability in real-world agricultural conditions.

No Explainability

Traditional classification models provide only disease labels without reasoning, severity assessment, or treatment recommendations.

Delayed Diagnosis

Manual inspection is time-consuming, inconsistent, and difficult to scale across large agricultural regions.

Hallucinations & Misclassification

Models often classify non-plant or unseen inputs with high confidence, reducing trust and reliability.

Poor Scalability

Single-crop and closed-set systems struggle to generalize across multiple crops, diseases, and field conditions.

⚠ **Key Challenges:** Dataset imbalance • Field image variability • Out-of-distribution detection • Multi-crop generalization • Real-world deployment constraints

Project Objectives

Build Preprocessing Pipeline

DDesign augmentation, normalization, and image quality workflows.

Create Crop Validation System

Develop pixel-level and embedding-based validation to filter non-plant images

Train Disease Classification Models

Develop and evaluate CNN-based disease classification systems.

Explore Transfer Learning Architectures

Evaluate EfficientNetB3, MobileNetV2, and ConvNeXt-Tiny.

Explore Vision Language Models

Integrate Vision Language Models for caption-based, explainable diagnosis

Test Mahavistar Integration

Evaluate deployment feasibility and application testing.

Improve Model Robustness

Increase reliability under real-world conditions.

Create Scalable Solution

Support extension toward additional crops and diseases.

Expected Benefits of the Project

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Accurate Diagnosis

High-confidence disease identification reduces false positives and treatment errors

Better Productivity

Early detection preserves yield and reduces crop loss across growing seasons

Economic Benefits

Reduced input waste and targeted treatment lower costs for smallholder farmers

Time Savings

Real-time diagnosis replaces days-long manual inspection cycles

Scalability

Modular architecture supports new crops, regions, and languages without retraining

Real-Time Decisions

Field-deployable inference enables immediate farmer action at point of capture

Digital Agriculture

Bridges the gap between traditional farming and modern AI-driven agronomy

Sustainable Farming

Precision treatment reduces overuse of pesticides and promotes eco-friendly practices

> *"Empowering farmers with accessible and intelligent diagnosis tools"*

Project Milestones

From Classical Vision to VLM

The project progressed through four distinct phases, each building on prior learnings and expanding system capability.

Phase 1: Image Processing

RGB/BGR filtering, Sobel & Canny edge detection, resizing fundamentals

1

Phase 2: Preprocessing Pipelines

Augmentation, normalization, quality checks, and dataset curation

2

Phase 3: CNN Models

Custom CNN, EfficientNetB3, MobileNetV2, ConvNeXt Tiny — trained and benchmarked

3

Phase 4: Validation Systems

HSV pixel validation, MobileNet embeddings, accept/reject gating

4

Phase 5: LLaVA Attempts

LLaVA 1.5 & 1.6 — initial VLM exploration; identified hallucination issues

5

Phase 6: BLIP-2 & Florence-2

Alternative architectures tested; refined caption generation approach

6

Phase 7: Qwen2.5-VL

Selected production VLM with LoRA fine-tuning for structured diagnosis

7

Phase 8: Future Direction

Explainable diagnosis and caption-based systems.

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Technical Achievements for the Review Period

Preprocessing, Validation & CNN Models

Image Processing & Preprocessing

- **RGB/BGR Filtering** — Color space normalization for consistent input
- **Sobel & Canny** — Edge detection for structural feature extraction
- **Resizing** — Standardized input dimensions across all models
- **Augmentation** — Rotation, flip, zoom for dataset diversity
- **Normalization** — Pixel value scaling for training stability
- **Quality Checks** — Blur detection and artifact filtering

CNN Model Performance Comparison

Model	Train Accuracy	Validation Accuracy
EfficientNetB3	90%	93%
MobileNetV2	91%	90%
ConvNeXt-Tiny	98.80%	96.10%

Validation Pipeline

Pixel Validation:

- HSV color space conversion
- Green ratio thresholding
- Edge density measurement

Embedding Validation:

- MobileNet feature embeddings
- Cosine similarity thresholds

Output:

Accept / Reject gate before inference

Vision Language Model Exploration & Proposed Architecture

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VLM Model Evaluation

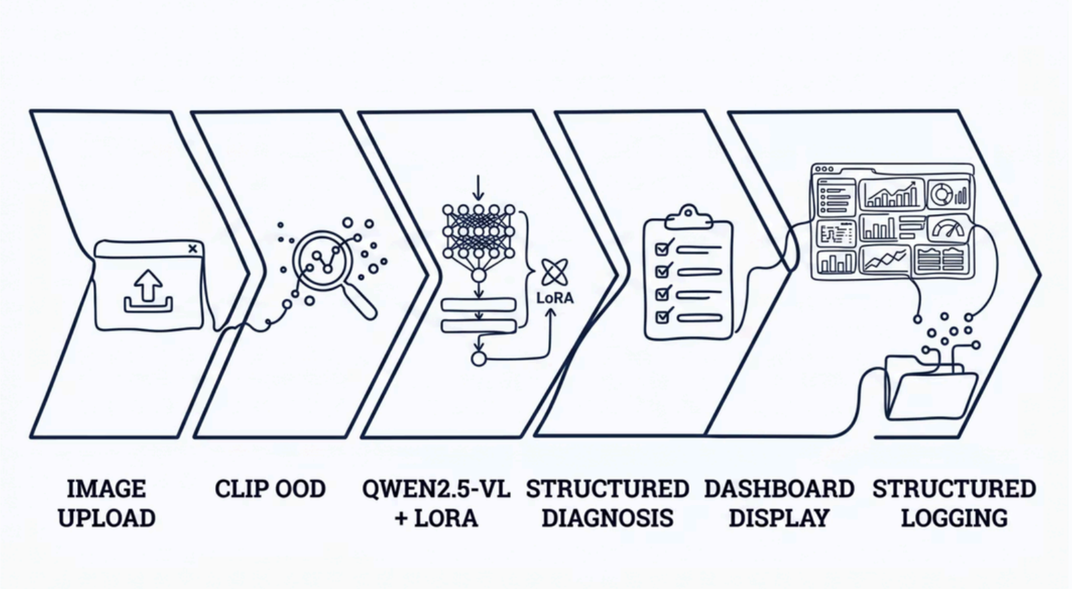
Model	Attempt Outcome	Key Learning
LLaVA 1.5	Hallucination issues observed during experimentation	Need stronger grounding mechanisms
LLaVA 1.6	Improved performance but remained unstable	Prompt engineering is critical
BLIP-2	Good caption generation quality	Limited disease-specific understanding
Florence-2	Promising structured outputs	Compute-intensive training requirements
Qwen2.5-VL	Most promising results during experimentation	Better explainability and structured outputs

Why Move Beyond CNNs?

- Limited explainability — no natural language output
- Single-label classification only
- No severity or confidence prediction
- Poor robustness on OOD images
- Hallucinations in ambiguous cases

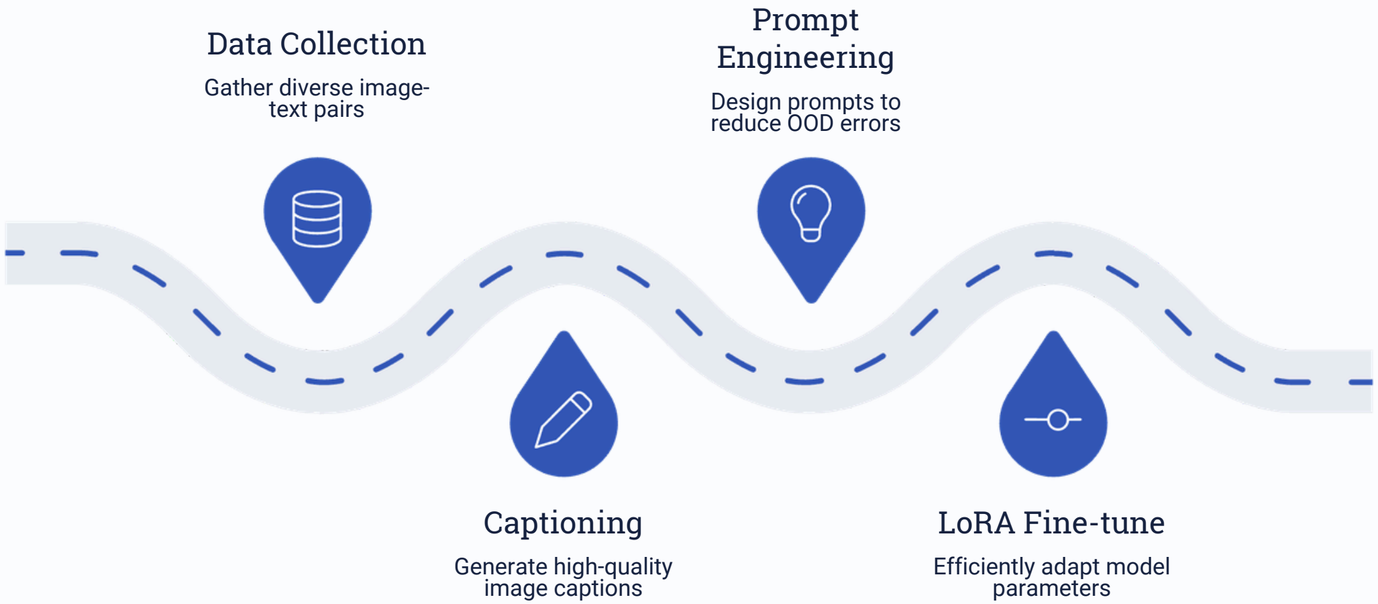
Transition Driver: VLMs provide caption-based, explainable, multi-attribute diagnosis — a qualitative leap beyond classification-only CNNs.

Production System Architecture



Training Pipeline, Engineering Challenges & Results

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Each stage was iteratively refined to address engineering challenges including batch mismatches, label masking errors, OOD hallucinations, resume failures, tensor dimension mismatches, and caption memorization — with targeted fixes applied at each step.

Key Performance Metrics

98.5%
Accuracy
Overall classification accuracy on validation set

0.6047
BLEU-1
Caption generation quality score

0.9654
Severity R^2
Disease severity prediction model fit

0.9827
Macro F1
Balanced precision and recall across all disease classes

0.5869
ROUGE-L
Longest common subsequence match for diagnosis text

0.025
Severity MAE
Mean absolute error in severity estimation

Shortfalls, Future Work & Conclusion

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Current Shortfalls

Compute Limitations

GPU constraints limit batch sizes and VLM training speed

Dataset Limitations

Limited crop diversity, class imbalance, and fewer real-world field images affect model generalization.

Long Training Cycles

Training and experimentation with larger architectures require significant time and computational resources.

Integration Complexity

Integrating experimental VLM pipelines with existing workflows requires further validation and testing.

Future Work Roadmap

- Expand dataset diversity and real-world image collection
- Strengthen crop validation and OOD detection mechanisms
- Continue exploration of Vision Language Models for explainable diagnosis
- Extend support to additional crops and disease categories
- Improve robustness and generalization across field conditions
- Introduce multilingual diagnostic outputs for accessibility

Key Takeaways

Explainable AI

Investigated Vision Language Models for more interpretable disease diagnosis.

Robust Validation

Two-stage pixel + embedding gating filters non-plant inputs

Deployment Testing

Disease models were tested within the Mahavistar ecosystem for usability evaluation.

Model Evaluation

Multiple architectures and evaluation metrics were explored to assess performance.

> *“From classical computer vision pipelines toward explainable multimodal systems for intelligent agriculture.”*

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Thank You